Year 12 Maths - Exponential models

Similar questions - Exponential functions and their graphs

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $y = e^{2x}$.

(a) Find the value of x for which $y = 7$, giving your answer to 3 significant figures. **[2]**

(b) Find the equation of the tangent to the curve at the point where $x = 1$. **[3]**

(c) Show that this tangent crosses the x-axis at $x = \frac{1}{2}$. **[1]**

(d) Show that, at every point of the curve, the gradient is equal to twice the value of y. **[2]**

2 The graphs of $y = 2^x$ and $y = 2^{x+3}$ are drawn on the same axes.

(a) Describe fully a single transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+3}$. **[2]**

(b) The point A lies on $y = 2^x$ and the point B lies on $y = 2^{x+3}$. The line AB is parallel to the y-axis and has length 21. Show that $2^x = 3$ and hence find the x-coordinate of A in the form $\log_2 k$. **[4]**

Question 1

(a) $e^{2x} = 7$, so $2x = \ln 7$ and $x = \frac{1}{2}\ln 7 = 0.973$

(b) $\dfrac{dy}{dx} = 2e^{2x}$, which is $2e^2$ at $x = 1$, where $y = e^2$. $y - e^2 = 2e^2(x-1)$, so $y = 2e^2x - e^2$

(c) $y = 0$: $2e^2x = e^2$, so $x = \frac{1}{2}$

(d) $\dfrac{dy}{dx} = 2e^{2x} = 2y$

Question 2

(a) $2^{x+3} = 8 \times 2^x$: a stretch in the y-direction with scale factor 8 (equivalently, a translation by $\binom{-3}{0}$)

(b) $2^{x+3} - 2^x = 8 \times 2^x - 2^x = 7 \times 2^x = 21$, so $2^x = 3$ and $x = \log_2 3$